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A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere. The Earth is a vibrant blue and green sphere against a dark, star-filled background. A white rectangular frame is superimposed over the center of the Earth.

COMP-2032
**Introduction to
Image Processing**

Lecture 4
Thresholding & Binary Images



Learning Outcomes

IDENTIFY

1. What is Thresholding
2. Adaptive Thresholding





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What is Thresholding?



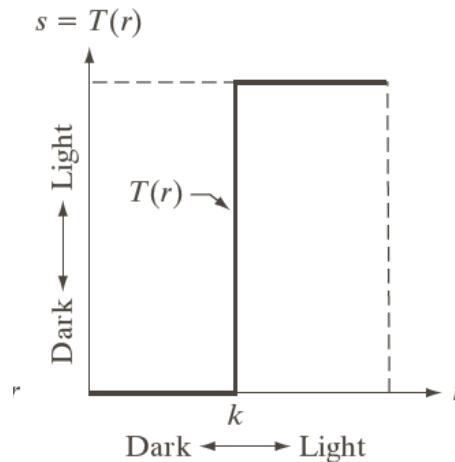
Binary Images

Many image processing operations make a decision:

Is this the colour/object/edge I am interested in?

The result is a binary image

- Pixels can have only **two values**: 0 or 1
- Binary images also need noise removal, enhancement, etc.



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Binarisation: Thresholding

- A dark object on a light background in a grey-level image
- Choose a threshold value, T
- Consider each pixel in turn

- If the brightness at a pixel is less than T , that pixel is object
- Otherwise, it is part of the background

Basic idea extends to colour; define sets of colour values that correspond to objects



Threshold, $T = 96$



Too Simple?

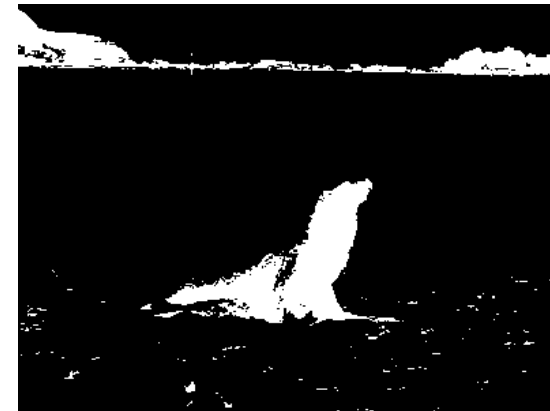
The value of the threshold is very important

- If it is **too high**, background pixels will be classified as **foreground**
- If it is **too low**, object pixels will be considered **background**

Assumes there are exactly two regions, with no overlap in their brightness – *is that true?*



$T = 128$

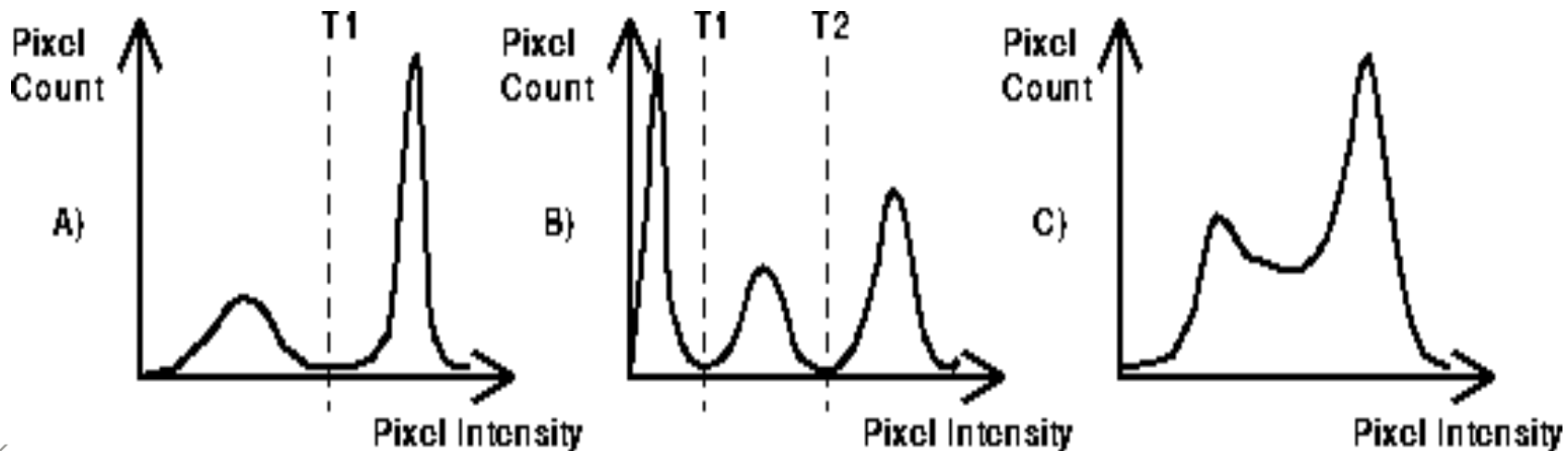


$T = 64$



Adaptive Thresholds

- If the user chooses t for each of a set of images, there is no guarantee the results will be consistent
- Automatic methods choose a threshold based on image properties: **histogram are commonly used**



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Adaptive Thresholding



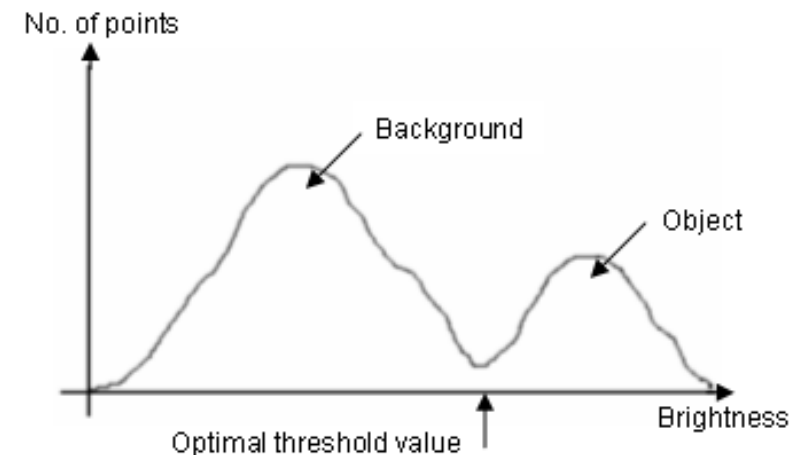
Otsu Thresholding

- Assumes histograms are *bimodal*; two regions can be separated by one threshold
- Think of the histogram as made of two normal (Gaussian) distributions, described by their means and deviations



If a threshold is wrong, it will include histogram bins from peak A in peak B

- Peak A's deviation will be too small
- Peak B's deviation will be too big
- Peak A's size (area) will be too small
- Peak B's size will be too big





Otsu Thresholding

- Find the threshold which minimises a weighted sum of the variations of the two regions that threshold produce
- Weights are the areas of the histogram assigned to each region

$$\sigma_w^2(t) = q_1(t)\sigma_1^2(t) + q_2(t)\sigma_2^2(t)$$

$$q_1(t) = \sum_{i=1}^t P(i) \qquad q_2(t) = \sum_{i=t+1}^I P(i)$$

This is small when the two regions are **both physically small and have low deviations**

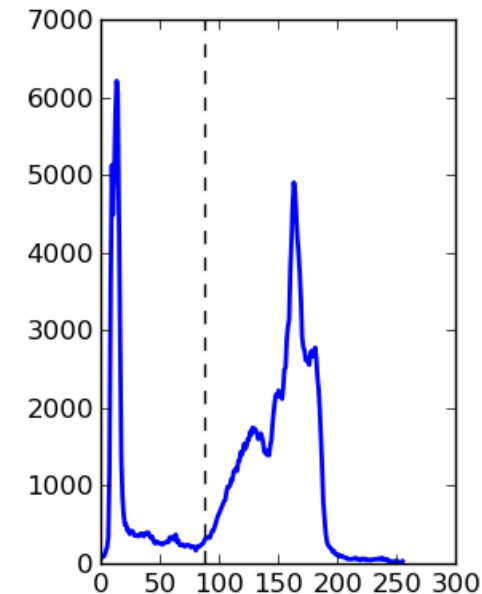


Otsu Thresholding

- Consider all possible threshold values (0 – 255)
- Compute weighted sum
- Pick t with the smallest value

Algorithm

A recursive version exists that is very efficient



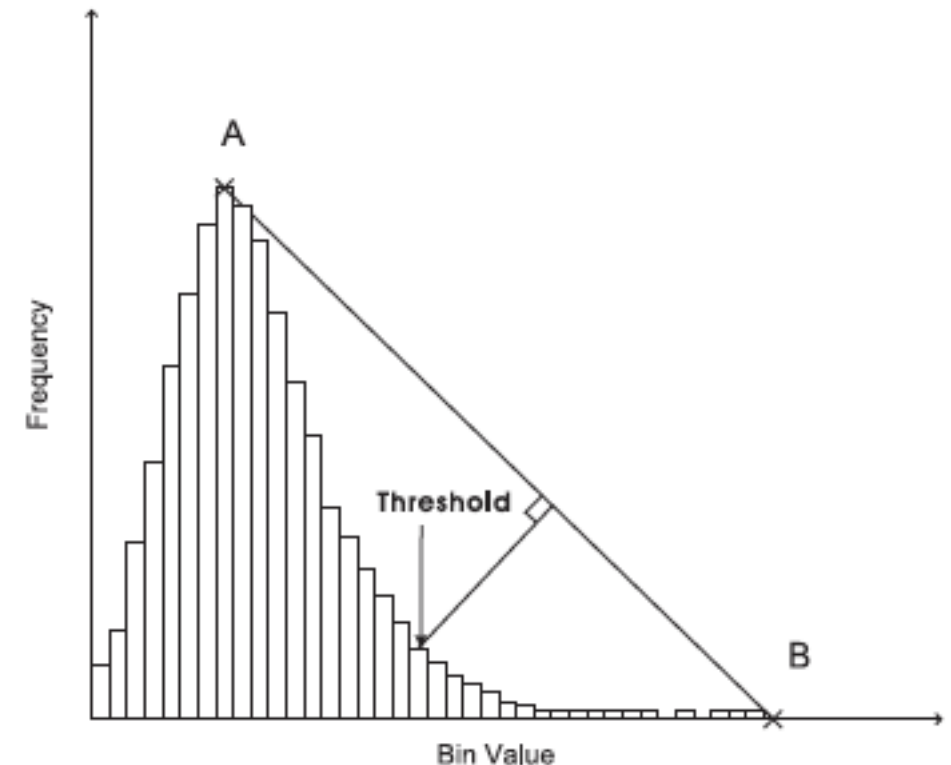


Unimodal Thresholding

Many histograms are not bimodal, there is often only one peak e.g., text is mainly white, with a small amount of black

- Finds the peak
- Draws a line from there to the top of the furthest bin
- Finds the top of the bin that is furthest from this line; that bin value is the threshold

Rosin's unimodal method



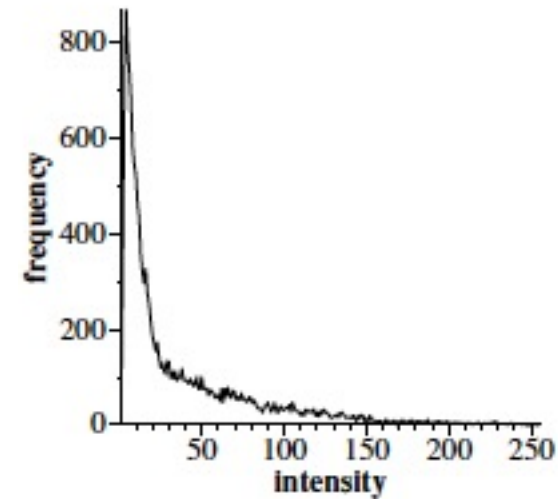


Unimodal Thresholding

Can be applied to any suitable image, e.g., [intensity gradients](#)



Edge of magnitudes
(inverted for display)



Histogram of edge
magnitudes



Unimodal Thresholding



Rosin

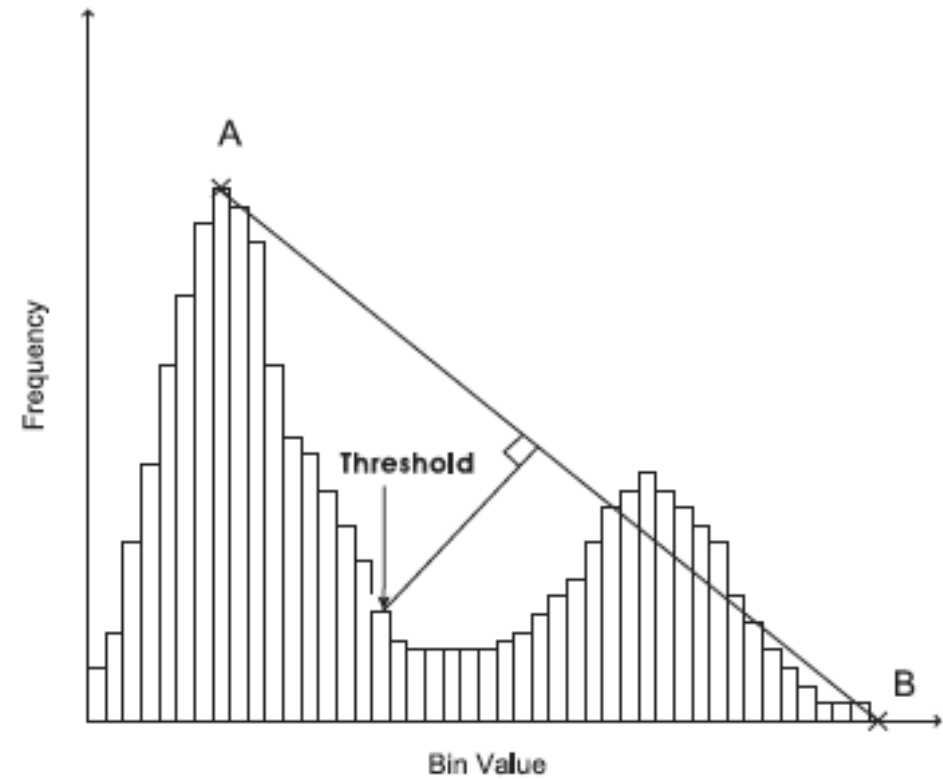
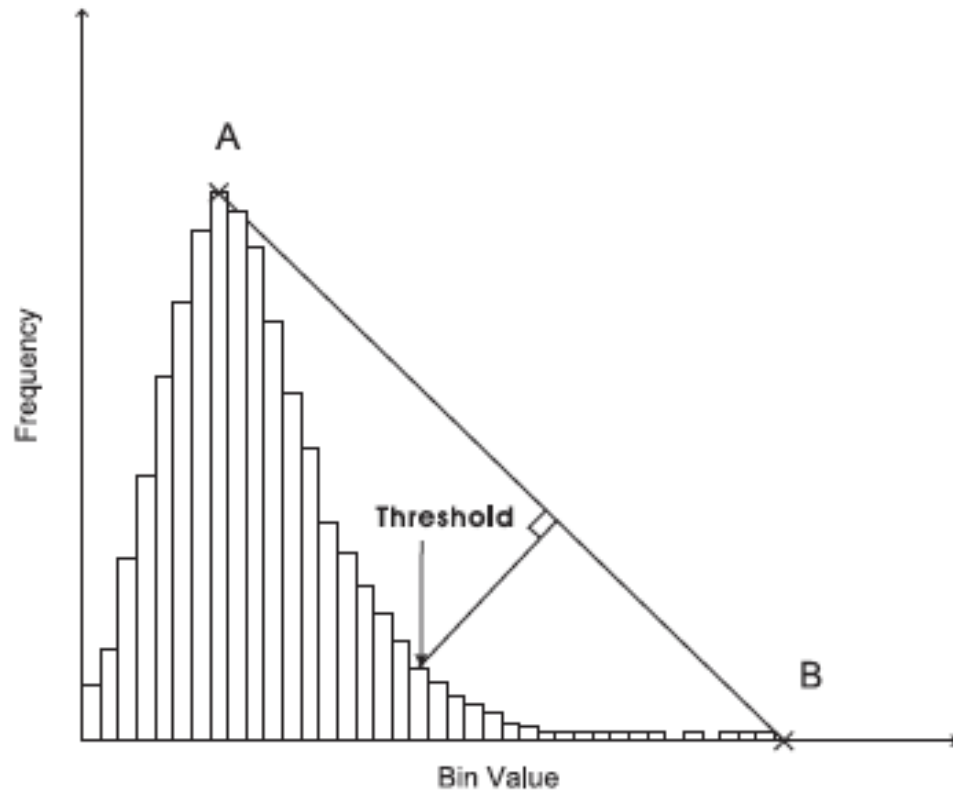


Otsu

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Otsu vs Rosin

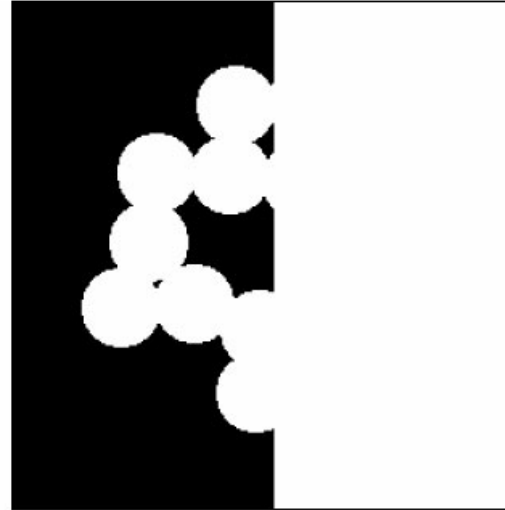
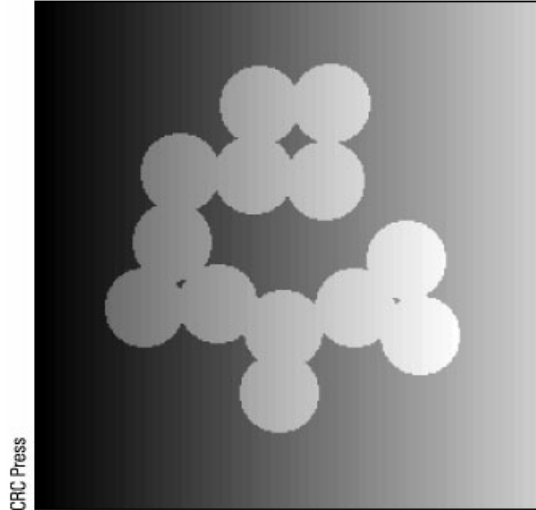


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Local Adaptive Methods

- Imaging conditions and object properties can vary within a single image as well as across sets of images
- Histograms can be too complex for any method's assumption to be true

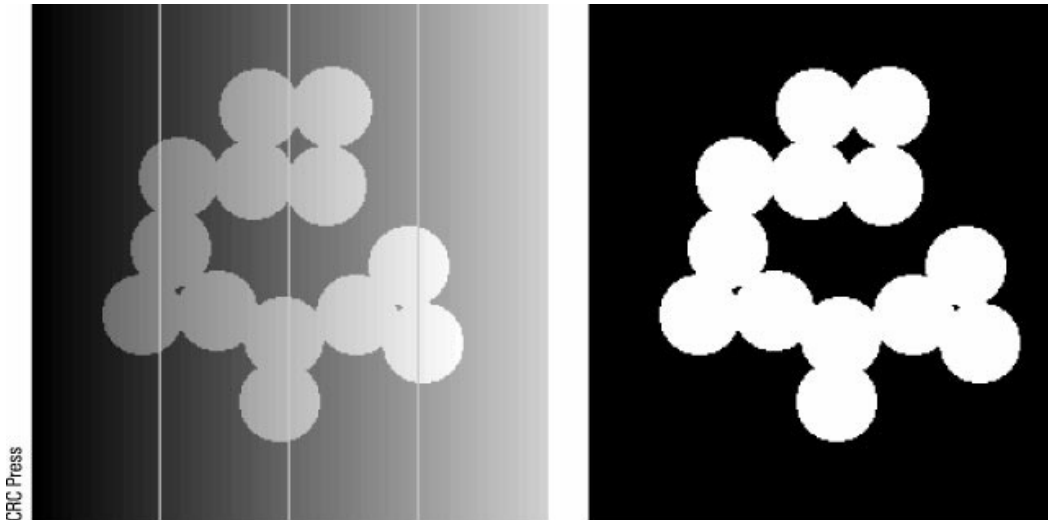


Otsu



Local Adaptive Methods

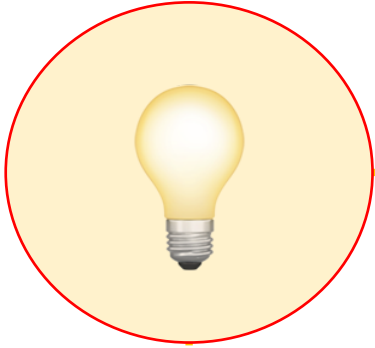
- Assumptions about histograms may, however, be true for local areas of the image
- Divide image into sub-regions, apply a threshold selection method independently to each



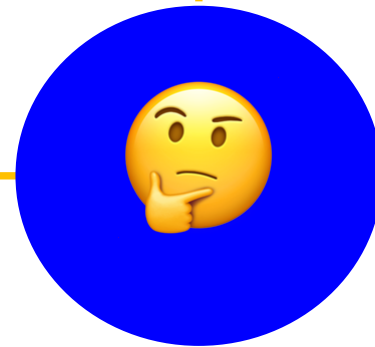
- The histograms of each vertical strip of this image are bimodal
- Otsu can be applied to each strip



Summary



1. What is Thresholding
2. Adaptive Thresholding





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Questions



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NEXT:

**Morphological
Operations**